

Goran Mržljak

Senior software developer

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Profile

I'm looking for hands-on senior development work: frontend, backend, or full-stack.

I can build your application from start to finish or join an existing team. I can define the technical architecture, help turn requirements into a working product, and support the team through mentoring, onboarding, and developer interviews.

- 18 years of professional experience. 10 years of contract work.
- I'm comfortable with both AI agent-assisted development and writing code without AI assistance.
- B2B contracts for full-time, part-time, or fixed-price work. Remote, with hybrid work possible in Zagreb.

Core technologies

Development React, TypeScript, Angular, Node.js, NestJS, PostgreSQL

Tools Jest, Docker, Git, Codex CLI, Copilot

Work experience

Independent contractor

January 2016 – Present

Senior software developer

Remote

HVAC monitoring and control

React, TypeScript, MobX, Tailwind CSS, FIN Framework, Project Haystack

An application for monitoring and controlling HVAC equipment across buildings and sites, developed for a large multinational. It formed part of a hardware and software solution for facilities such as hotels and office buildings.

- I joined an established team and built frontend sections, components, and controls to match UI designs. I fixed issues as they arose and worked with the specialist domain, technologies, and tools used by the platform.

Warehouse robotics

Angular, TypeScript, RxJS, Docker, PostgreSQL

A web application for managing autonomous warehouse robots and fleets. The wider system covered automation, planning, robot orchestration, and computer vision. Several related frontend repositories each contained about 100,000 lines of code, much of it shared across versions.

- I worked almost exclusively on the frontend, as one of two frontend developers in a project involving several dozen people.

- I independently migrated Angular 8 to 18, resolving over a thousand errors and refactoring libraries where necessary. I completed the migration without observed regressions, removing a major blocker for the security audit.
- I organized the changes for reuse across related robot and fleet applications and branches. I also implemented client requests, including dock monitoring and truck reception interfaces, pallet stacking support, and other robot operations.

Oracle Eloqua messaging integration

React, TypeScript, NestJS, MySQL, Prisma, Jest, OAuth, JWT, Nx

An integration connecting Oracle Eloqua to a mobile messaging service, allowing users to send SMS and WhatsApp messages within marketing campaigns. An embedded React interface supported configuration and monitoring.

- I independently built the frontend, backend, database, and external API integrations. Within platform constraints, I made technical and architectural decisions, clarified requirements with the client, and demonstrated progress.
- The roughly 20,000-line application coordinated two-way asynchronous communication. Campaign flows involved hundreds of thousands of messages, callbacks, and delivery reports. I coordinated concurrent operations and prevented race conditions so campaigns could progress without losing messages.
- Load testing exposed race conditions that I fixed and covered with regression tests. Backend coverage included unit tests, every endpoint, and complex asynchronous flows. Mockable database and service interfaces kept tests fast, and detailed logging supported diagnosis.
- I implemented OAuth 2.0 for requests to Oracle Eloqua, OAuth 1.0a for requests from Eloqua, API keys for the messaging platform, and JWT between the embedded frontend and backend.

Electricity consumption mobile application

React Native, TypeScript, Expo, Jest

A mobile application for the Norwegian public to monitor consumption, choose electricity providers, and view statistics and insights.

- I worked exclusively on the React Native frontend in a team of eight developers, implementing screens and features.

Mobile fueling operations

Node.js, Express, MongoDB, Angular, TypeScript, NgRx, Mocha, Nx

A production system supporting the operations of a leading mobile fueling company in California. The company had several hundred employees, with dozens of developers working on its web applications, mobile apps, and shared backend.

- I spent about 70% of my time on the Node.js backend, from REST APIs to the database, and 30% on the Angular administration app.
- I owned parts of the system and delivered changes affecting the whole application. Careful planning with developers and project managers allowed these changes to proceed without significant issues.
- I ran production MongoDB migrations using JavaScript scripts. Careful testing and execution kept those migrations error-free during my time at the company.

- I drove improvements to testing practices and the introduction of a monorepo where practical. I also interviewed, onboarded, and mentored developers.

Insurance platform

React, TypeScript, Redux, RxJS, Storybook, Material UI, Jest, Cypress

A platform for a UK client connecting parties in the insurance process and supporting searches for insurance schemes. The frontend contained about 100,000 lines of code.

- I joined after a year and a half of development and worked exclusively on the frontend, mostly as its sole developer.
- I refactored application sections, fixed typing issues throughout the codebase, rebuilt state management and asynchronous API handling, and reorganized the frontend architecture.
- I improved the build and development tooling, introduced linting and formatting, and added tests. Utility code had good coverage, with basic component coverage and a handful of Cypress tests for larger flows.
- I built a Storybook component library around Material UI that matched the existing design, while continuing feature development and bug fixes.

Accounting and transactions

Angular, TypeScript, RxJS, Jest, AG Grid

A large accounting application for a Finnish company, with dozens of screens containing complex business logic, forms, and data viewing and editing features.

- A colleague and I joined after limited prototype development and did most of the initial frontend implementation, with room to make architectural decisions.
- As the frontend team grew to more than ten developers, I implemented application-wide changes, reusable subsystems and patterns, larger features and screens, and maintenance work.

Graveyard management

Vue, TypeScript, OpenLayers, Kotlin, Spring Boot, PostgreSQL, Liquibase, Jest

A multi-tenant application for managing graveyards, graves, and information about the deceased and grave owners, with interactive maps and geospatial data for each grave.

- I implemented at least 95% of the application. The frontend used Vue, Vuex, Vuetify, and OpenLayers, with Jest unit tests.
- I built the Kotlin/Spring Boot backend, using Liquibase for schema changes and migrations. PostgreSQL served production, while an in-memory H2 database supported development and testing. Jasper Reports produced PDF reports.
- I wrote backend integration tests in Jest and TypeScript, exercising the full backend stack and much of its functionality against H2.
- I used Docker and Docker Swarm in development for automation and deployment experiments. Production deployment remained partly manual, with scripts handling most steps.

Parking payments backend

Java, REST APIs

A public mobile application for buying parking tickets for lots and garages in Zagreb.

- I built the central message hub connecting system endpoints and forming the top layer of the backend. It exposed the REST API for Android and iOS, validated requests, and integrated directly with parking provider APIs.
- It connected to the data access layer and to payment providers through another developer's component. My work was on the backend, not the mobile apps.

National single sign-on integration

Java, JSP, SOAP

An integration with NIAS, Croatia's National Identification and Authentication System, for the Croatian Postal Service.

- I implemented the entire integration. It allowed existing NIAS users to sign in to the Postal Service's system and enabled the Postal Service to act as a credentials provider for other NIAS services.

Government payments validation

Java, JUnit, JSON

A reusable component validating payments to the Croatian government, agencies, and local and regional administrations. It checked individual invoice fields and combinations against configurable rules, returning structured errors and warnings.

- I implemented the component and extensive unit tests in less than two weeks. Requirements came from about 180 pages of rules, supporting tables, and data.
- I translated the rules into a JSON configuration of about 3,000 lines and wrote the logic to parse and apply them.
- QA found about three failures in 100 manual scenarios. I converted their spreadsheet results to CSV-backed automated tests, fixed the failures, and retained every scenario as regression coverage. No further issues were reported while I worked for the client.

EU roaming usage and traffic tracking

Java, Spring Boot, JPA/Hibernate, Querydsl, Liquibase, Jasper Reports

A system for Croatia's largest telecom, providing roaming usage statistics and helping detect abuse during changes to EU roaming rules.

- I implemented the entire backend for its administration application, from the REST API to the repository layer, and built its reporting system with Jasper Reports.

E-learning MVP

AngularJS, Java, Spring Boot, PostgreSQL, Video.js, Veeting Rooms

An online learning MVP with video courses and online meetings with whiteboards.

- After initial work by other developers, I implemented the rest of the application. I also handled deployment, communicated with clients, and helped define requirements.

Earlier employment

APIS IT

November 2013 – May 2016

C# Developer

Zagreb (Croatia)

e-Tax platform

C#, ASP.NET Web Forms, jQuery, Microsoft Reports (.rdlc), IBM DB2

A large application with more than one million daily views, allowing Croatian citizens and businesses to submit tax forms electronically. It supported many types of tax reporting alongside other features, with form submission probably its main use case.

- I fully implemented seven tax forms in production as of March 2016 and contributed to about ten others, from bug fixes to substantial implementation. The system contained about 25 forms at the time.
- Each form required a web interface, XML import and export, PDF reporting, extensive server-side validation against business and legal rules, and changes to the data access layer.

Location and building permits

C#, ASP.NET MVC, jQuery, Fluent Validation, Aspose.Words, Oracle Database

A relatively simple public application for submitting data for location and building permits, built for Croatia's Ministry of Construction and Physical Planning.

- I implemented much of the UI, including views, models, and controllers, a substantial part of the business logic, the entire data access layer, and PDF reporting.

Gemmeus d.o.o.

October 2012 – January 2013

C# Developer

Remote

Spreadsheet component

C#, Web Forms, WPF, Windows Forms

A component for programmatic access to Microsoft Excel spreadsheets, supporting reading, writing, data manipulation, conversion, and printing.

- I handled various development tasks, including exporting Excel data to a Windows Forms DataGrid while matching the original content, formatting, and styles as closely as possible.

INETEC d.o.o.

November 2009 – October 2012

C# Windows Desktop (WPF) Developer

Zagreb (Croatia)

Nuclear inspection and damage analysis

C#, WPF, Cinch (MVVM framework)

A large WPF desktop application for operators to review inspection data and analyze damage in nuclear power plant steam generators.

- I developed a large part of the application.
- I created custom WPF controls for 2D and 3D signal charts and plots. Rendering ran on a separate thread to keep the application responsive during intensive data processing and display.

- I worked on data persistence and versioning, signal filtering and mixing, and many WPF views and subsystems, from the UI to the persistence layer.

Inspection robot control

C#, WPF, Cinch (MVVM framework)

A prototype robot for positioning inspection probes could walk along a steam generator tube wall, reducing or eliminating operator exposure to ionizing radiation.

The WPF desktop application controlled the robot, displayed its position, monitored its state, and complemented hardware and electronic handling of edge cases and failures.

- I implemented most of the application, including the entire GUI. It consisted mainly of custom graphical controls, with multithreading for the more complex displays to keep the application responsive.
- I built or fully rewrote nearly all the backend code down to the electronics controller interface.
- Another developer wrote the original controller code in a C-like language. I maintained it and made the required changes and fixes.

Ultra d.o.o.

October 2008 – November 2009

C# Windows Desktop and Web Developer

Karlovac (Croatia)

Fuel dispenser web application

C#, ASP.NET Web Forms, PostgreSQL

A web application for configuring and monitoring fuel dispenser drivers at fuel stations.

- I implemented substantial parts of the application and maintained the entire project.

Fuel dispenser desktop application

C#, Windows Forms, SQLite

A desktop application with a similar purpose to the web application, serving different drivers and clients.

- I built the entire application, with custom controls for displaying and controlling fuel dispenser state and SQLite for storage.

Ekobit d.o.o.

February 2008 – October 2008

C# Web Developer

Zagreb (Croatia)

Financial web applications

C#, ASP.NET Web Forms, Microsoft SQL Server

Web applications for Croatian financial institutions, including banks, funds, and stock exchanges.

- I handled small development tasks and bug fixes in the applications and their databases.

Visage Technologies (Student Project)

C++ Developer

January 2007 – August 2007

Zagreb (Croatia)

Body animation exporter

C++, 3ds Max SDK

An exporter that converted body animations from 3ds Max Biped to MPEG-4 FBA for a larger face and body animation system.

- I implemented the entire exporter.

Education

Faculty of Electrical Engineering and Computing (FER), University of Zagreb

Graduate Engineer of Computing. Five-year degree program, totaling 284 ECTS credits.

For my thesis in computer graphics, I built a BVH animation viewer that parses motion capture files and renders the animation.

[Thesis summary on the faculty site](#) · [Thesis application source code](#)

Languages: Croatian - native · English - fluent

Skills

Extensive and Recent Experience

Programming Languages: TypeScript, JavaScript, HTML, CSS

Libraries and Frameworks: React, Angular, Express, NestJS, Tailwind CSS, Material UI, Redux, MobX, RxJS, NgRx

Platforms: Node.js, Bun

Tools: ESLint, Prettier, Vite, Webpack, Jest, Vitest, Storybook, Docker, Bash

Databases: PostgreSQL, MySQL, MongoDB

CI/CD: GitHub Actions

AI: Codex CLI, Copilot

Operating Systems: Windows, macOS, Linux

Extensive Experience, Not Recent

Programming Languages: Kotlin, Java, C#, C++

Libraries and Frameworks: Spring Boot, JPA/Hibernate, Liquibase, Vue, AngularJS, ASP.NET MVC, ASP.NET Web Forms, WPF, Windows Forms

Platforms: JVM, .NET Framework

Tools: React Testing Library, Mocha, Nx (monorepo), Prisma, Maven, JUnit

Databases: H2, SQLite, Microsoft SQL Server

Limited Experience

Programming Languages: Ruby, Python

Libraries and Frameworks: Android SDK, GraphQL

Tools: Playwright, Cypress

Databases: IBM DB2, Oracle Database

AI: Claude, Cursor